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Research on a hybrid prediction model for stock price based on long short-term memory and variational mode decomposition

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researchers have proposed many prediction methods. Many papers provide strong evidence that stock prices can be predicted from past price data. In this paper, we propose a hybrid prediction model for stock price based on long short-term memory (LSTM) and variational mode decomposition (VMD). We use the variational mode decomposition method to decompose the complex time series of stock prices into several relatively flat, regular, and stable subsequences. Then, we use each subsequence to train the long- and short-term memory neural network and predict each subsequence. Finally, we merge the predicted values of several subsequences to form the predicted results of the stock price complex original time series. To verify fully the method, we selected four experimental data for testing. Compared with the prediction results of various prediction methods, the prediction accuracy of our proposed model is higher. Especially in the R^2 index, the experimental effect is very good. The proposed method achieves good results of more than 0.991 on each data set. Therefore, our proposed hybrid prediction model is accurate and effective in forecasting stock prices and has practical significance and reference value.

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Data Availability

Data are fully available without restriction. The original experimental data can be downloaded from Yahoo Finance for free(<http://finance.yahoo.com>).

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Conflict of interest

The authors declare that they have no conflicts of interest.

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Keywords

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